

Manual

Failure Mode Analysis / Action Tracking WEP-FSM 8.1

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1 Failure Mode Analysis / Action Tracking

Purpose

This component is used to determine failure modes on the basis of collected failures in order to introduce improvements that enhance the productivity.

Action tracking enables the central processing of actions and to filter these for example for actions yet to be concluded.

Implementation Considerations

Use of this component is recommended when, within the collection of inspection data, failures are automatically generated or actions are taken through the AIP.

Integration

The component "Inspection Planning for Goods Receipt Inspections" with its associated inspection requirements and the failures collected with the inspection results forms the base data, where the inspection data must have been previously entered through the AIP data entry client.

Features

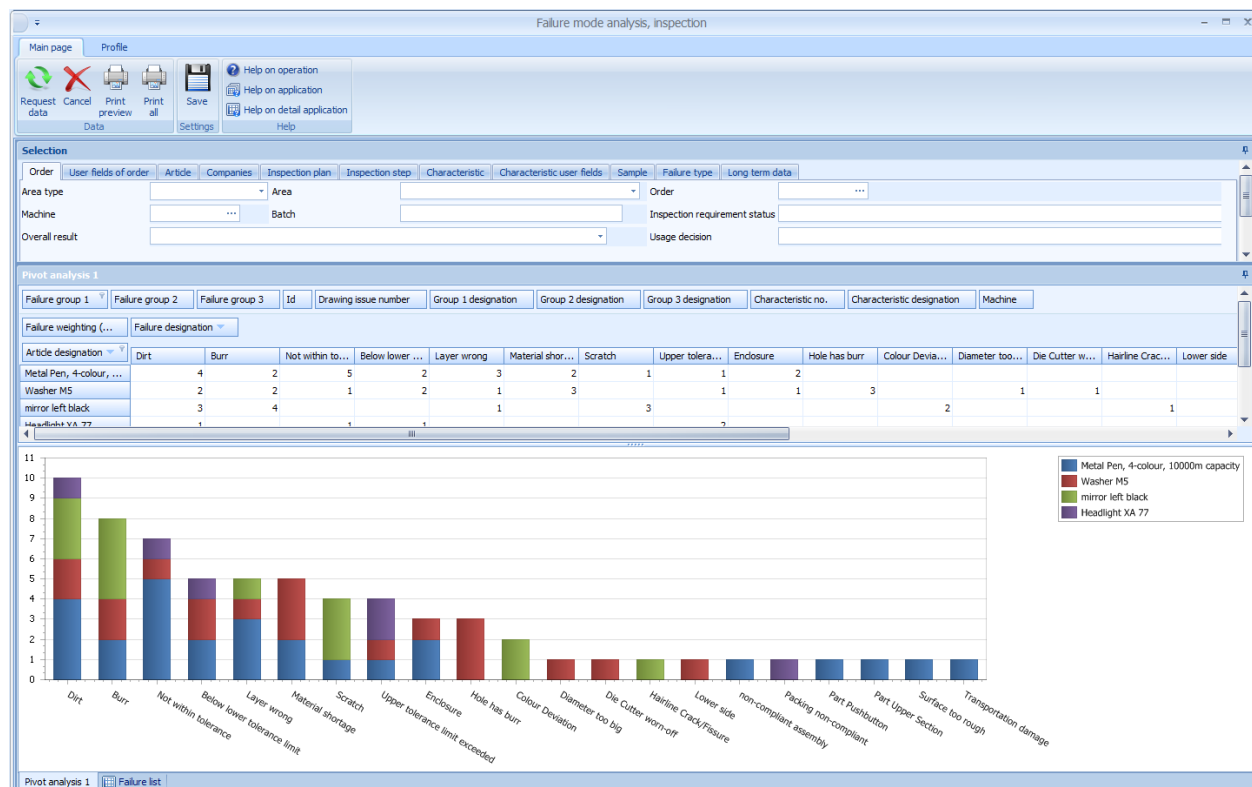
The component contains the following functions:

- Graphical pivot analysis evaluation of collected failure types, locations and causes with a wealth of filtering criteria and display options
- Cross-sectoral tracing of measures/action tracking including editing capabilities (status, effectiveness, degree of completion, actual date, comments on activities taken, filtering according to responsibilities etc.)

2 Failure Mode Analysis

Summary

Menu	Quality management → QM evaluation → Failure mode analysis
Transaction code	faep
Function authorization	faep



Utilization

The failure mode analysis allows for failures of the following types recorded during the inspection process or generated automatically to be evaluated.

- Failure type (FT)
- Failure location (FL) and
- Failure cause (FC).

In this context evaluations/reports are based on pivot functions. They allow for the distribution of failure types (frequency) to be presented for each article/item referring to a period of time that has been filtered before. These analyses help determine the core areas that might require action to be taken to optimize the production process with respect to quality.

Integration

The failure mode analysis evaluates failures of inspection requirements from the area types:

- goods receipt,
- initial sample inspection,
- production and
- goods issue

Consequently, the failure mode analysis is a comprehensive analysis tool.

Prerequisite

There are no special requirements. The only prerequisite is that failures have to be recorded, which, in turn, need to be defined within the master data of quality management before.

Selection criteria

The paragraph that follows shows some of the available selection criteria. Self-explanatory filter options are not listed.

"Order" tab

Area type

Selection list of area types

- in-production inspection
- goods receipt inspection
- goods issue inspection
- initial sample inspection

The list of area types may be restricted subject to the respective licenses in use.

Area

Selection list of the configured areas of the previously filtered area type. By default, the following areas are available. .

- In-production inspection: production
- Goods receipt inspection: goods receipt
- Goods issue inspection: goods issue
- Initial sample inspection: initial sample

Further areas can be generated through customizing.

Inspection request status

Selection list including multiple selection options for all inspection request statuses

Overall result

Selection list for the results of inspection requests (pass, fail, conditionally pass)

Usage decision

Selection list of usage decisions (e.g. release, conditional release, rework, reject, sort) that are available, once an inspection request has been completed.

"Inspection point identification" tab

If these fields are not available, you require a new program version of this application.

Workplace

Workplace number

Partial batch

Partial batch number

Batch

Batch number

Field 1...3

Additional fields for inspection points

"Additional fields for inspection points" tab

If these fields are not available, you require a new program version of this application.

Field 4...8

Additional fields for inspection points



The fields "field 1" to "field 8" are enabled subject to MPDV customizing and are assigned individual names. As these field names are flexible, they are only entitled "field 1" to "field 8" in this document.

Field 1 includes, for example, the tool if cavity-related data collection is enabled.

"Sample" tab

Machine sample

Selection list of machines/workplaces

The machine assigned to the first measured value is filtered here, provided that different machines are assigned to the measured values of the same sample.

Toolbar

There are no other special function buttons in addition to the standard functions/features.

"Graphic failure analysis" detail applications

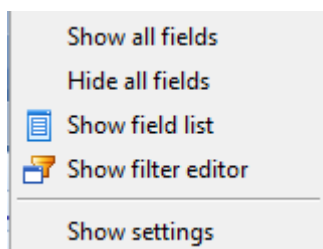
Data is displayed in a pivot table in combination with bar charts. Different application functions are provided for the presentation. The failures that have been filtered beforehand are the data basis.

The general pivot functions are not described in more detail in this document. The paragraphs that follow only describe the elementary functions of this evaluation/report.

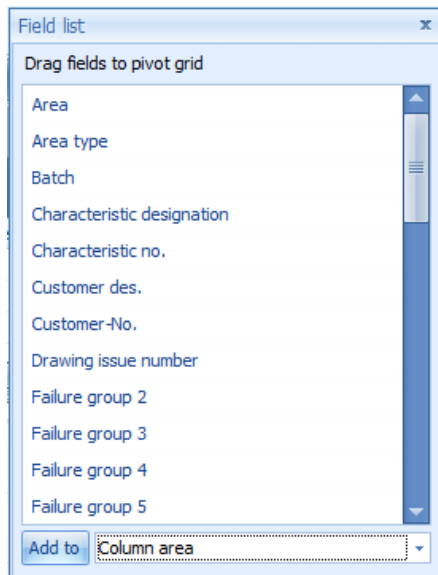
Pivot evaluations/reports provide the following benefits.

- Large amounts of data may quickly be summarized and presented.
- Rows and columns can be exchanged to have the source data summarized differently.
- Simple filters by "drag and drop" with additional detail filters.
- Due to an interactive way of representation, data can be summarized and analyzed in different formats and using different calculation methods.

The below context menu can be opened by clicking the right mouse button.

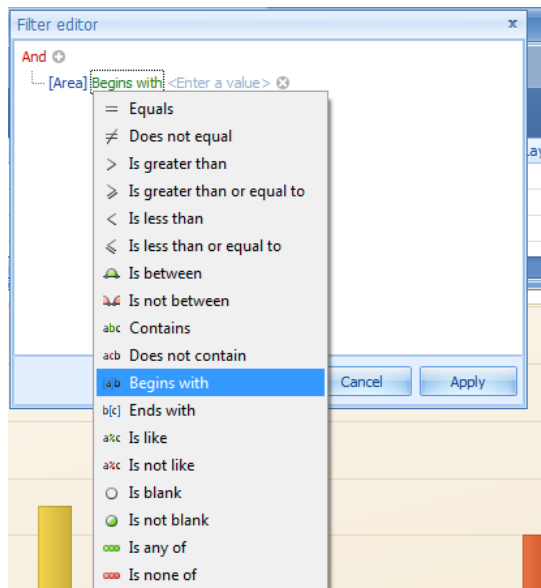


The function "show field list" allows for the fields that are to be used in the pivot analysis to be selected. The below figure shows a possible field list.

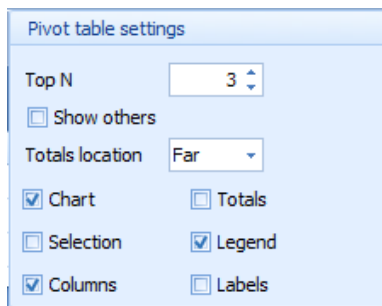


The requested fields may be put into the evaluation area by drag & drop.

In addition to the selection criteria, the "show filter editor" function enables further flexible restrictions of the data basis.

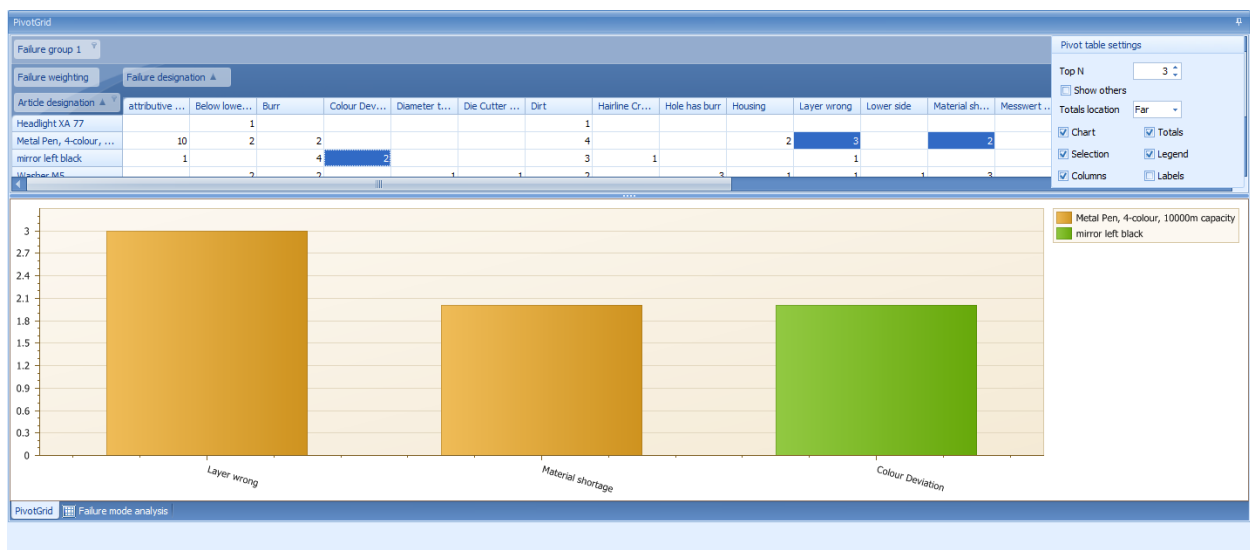


The below dialog is opened to show the settings made.



If the "selection" option is checked entire areas may be selected in the table view. In this case, the graphic representation is based on the selected rows. If the "label" option is checked it is possible to show the total number of each bar.

The below figure explains these functions.



The row showing the total result may be displayed additionally in the bar chart if the "totals" option is checked. Provided that the "selection" function is checked and the corresponding cells of the "total result" row are selected, the total result of the corresponding column is added to the respective bar.

It is switched between the graphic presentation of the corresponding number of columns or rows by checking/unchecking the "columns" option.

"Failure list" detail applications

The failure list shows the failures including the referenced data that are filtered based on the used selection criteria. Normally, the referenced data corresponds to the field list for the pivot analysis.